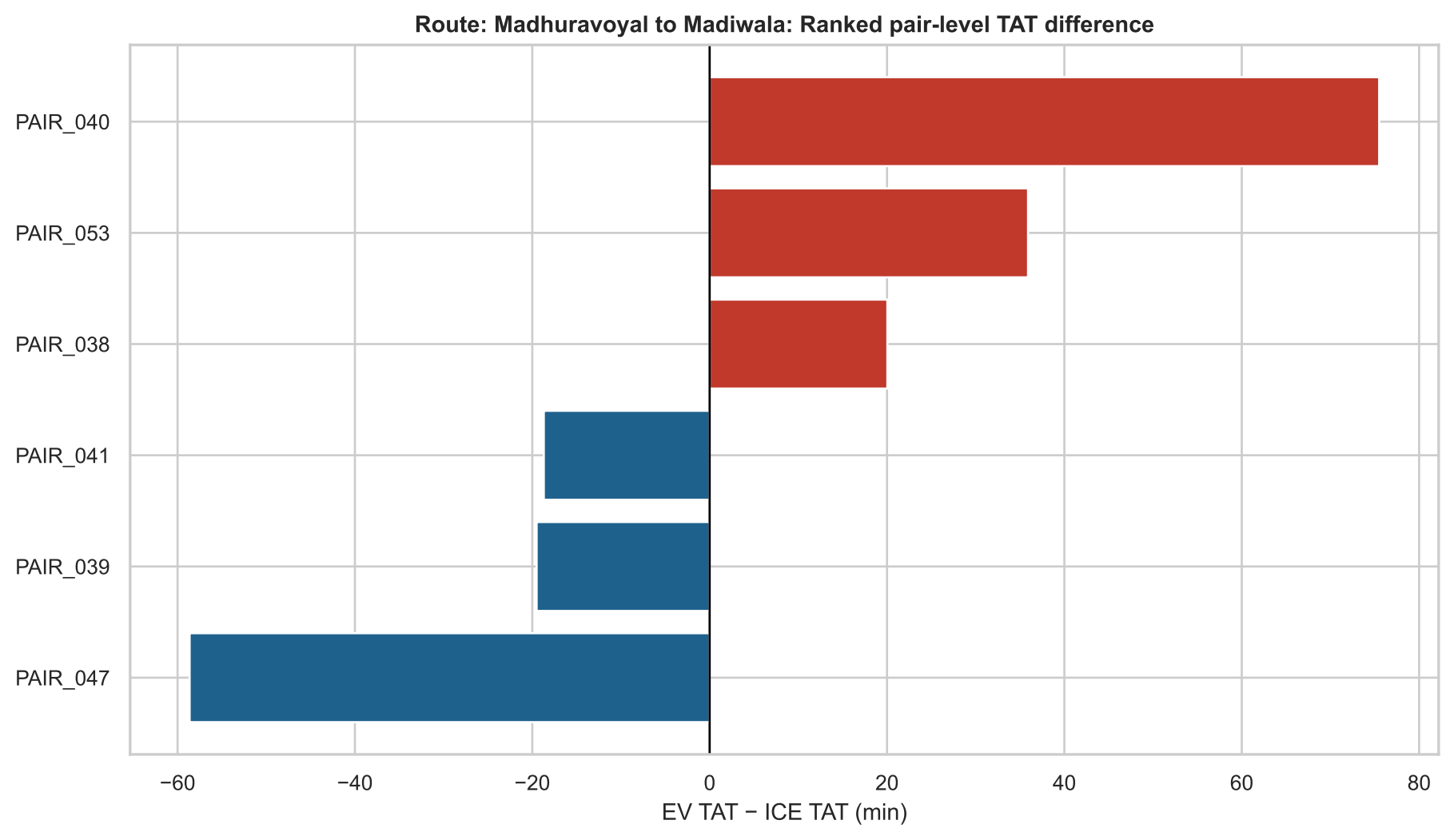
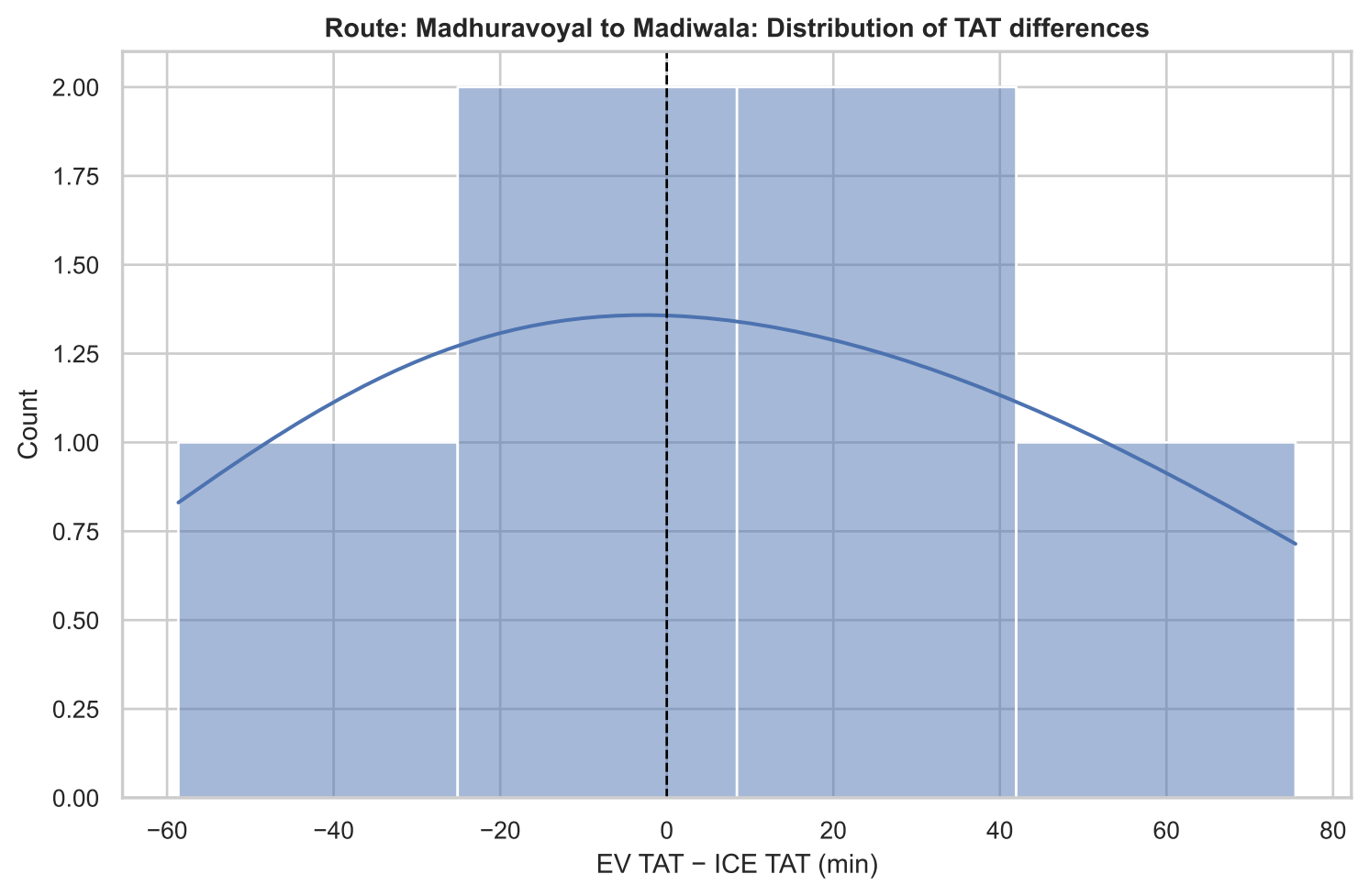
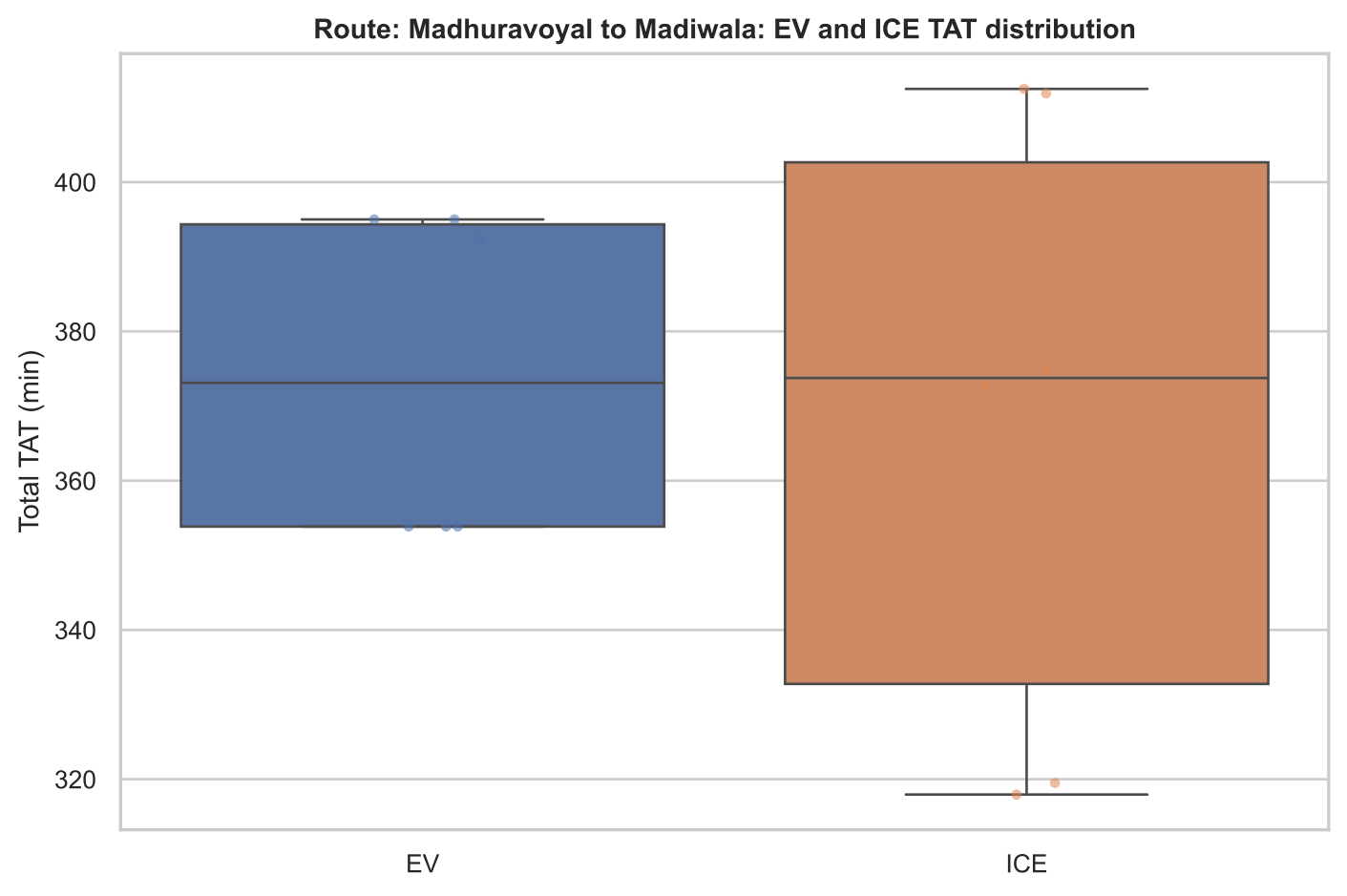
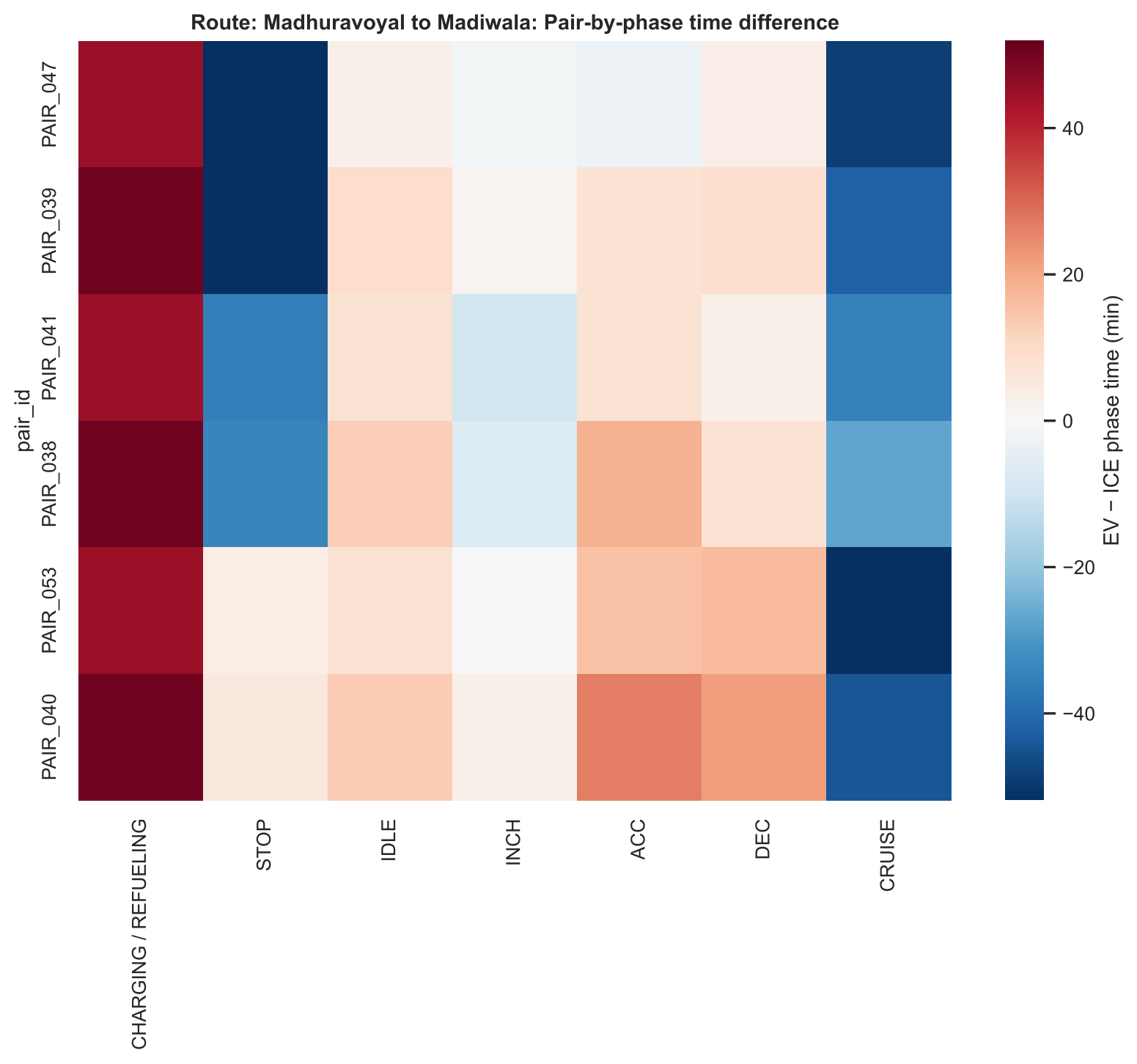
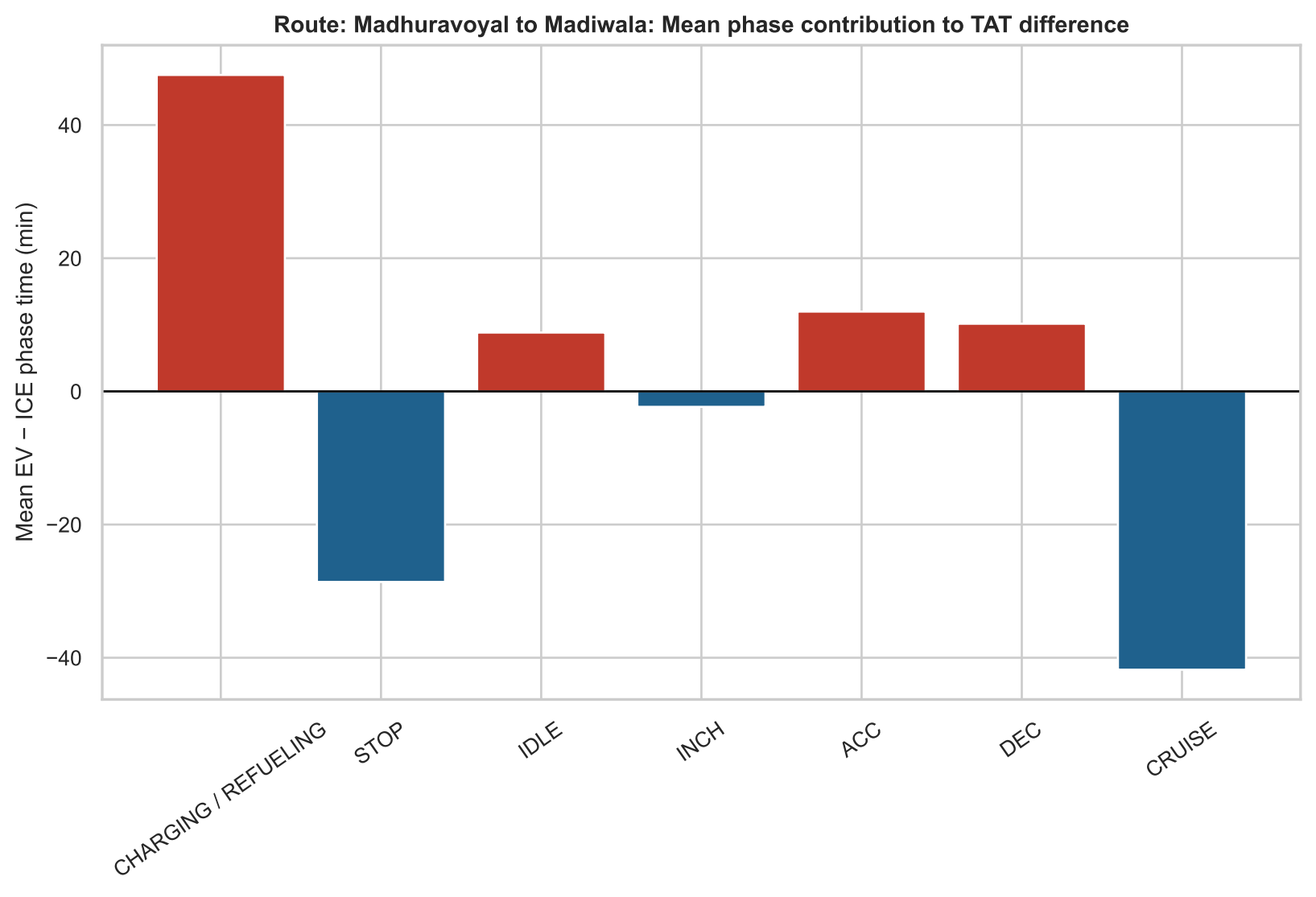
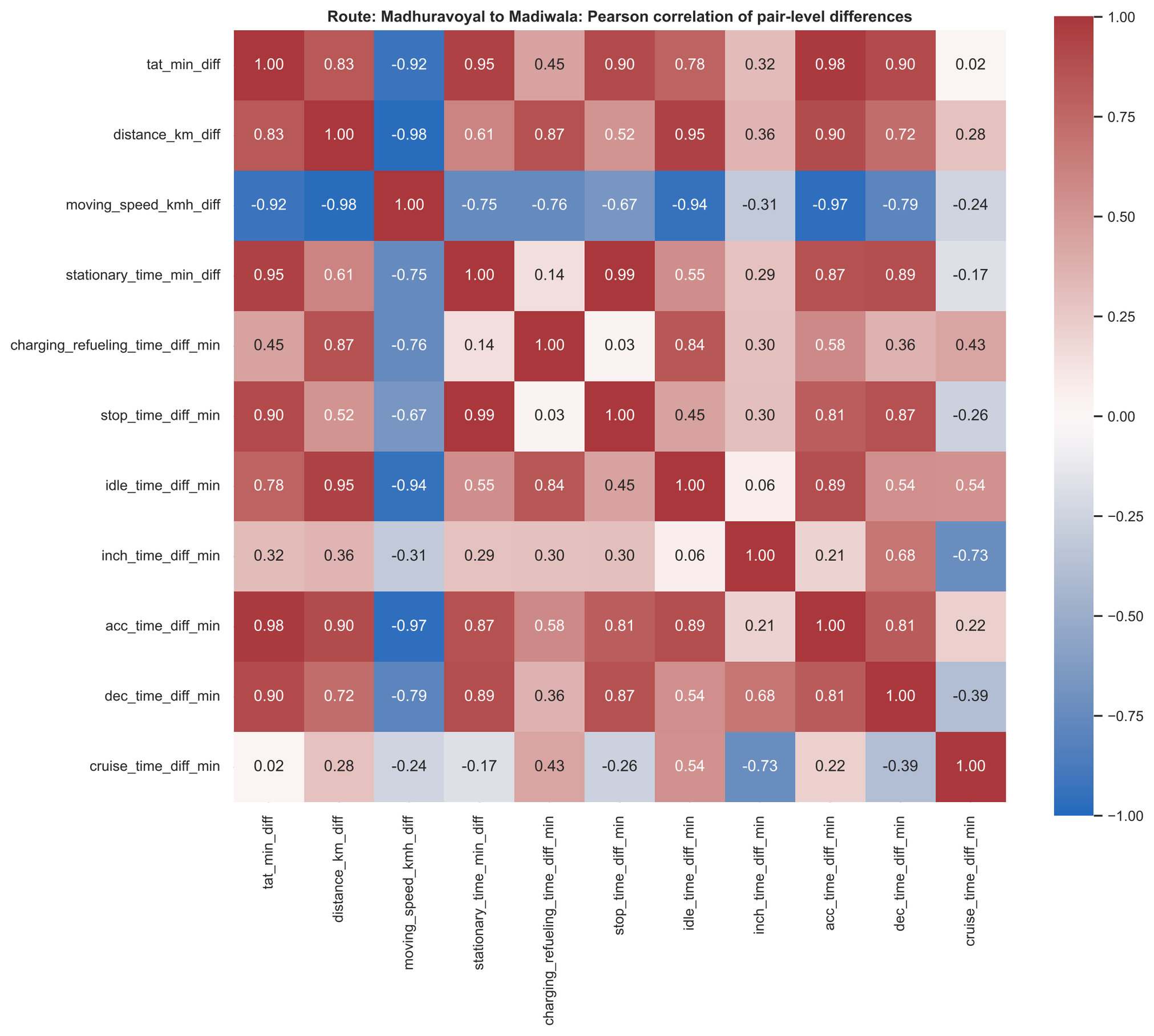

Route: Madhuravoyal to Madiwala: Ranked pair-level TAT difference
PAIR_040
PAIR_053
PAIR_038
PAIR_041
PAIR_039
PAIR_047
−60
−40
−20
0
20
40
60
80
EV TAT − ICE TAT (min)

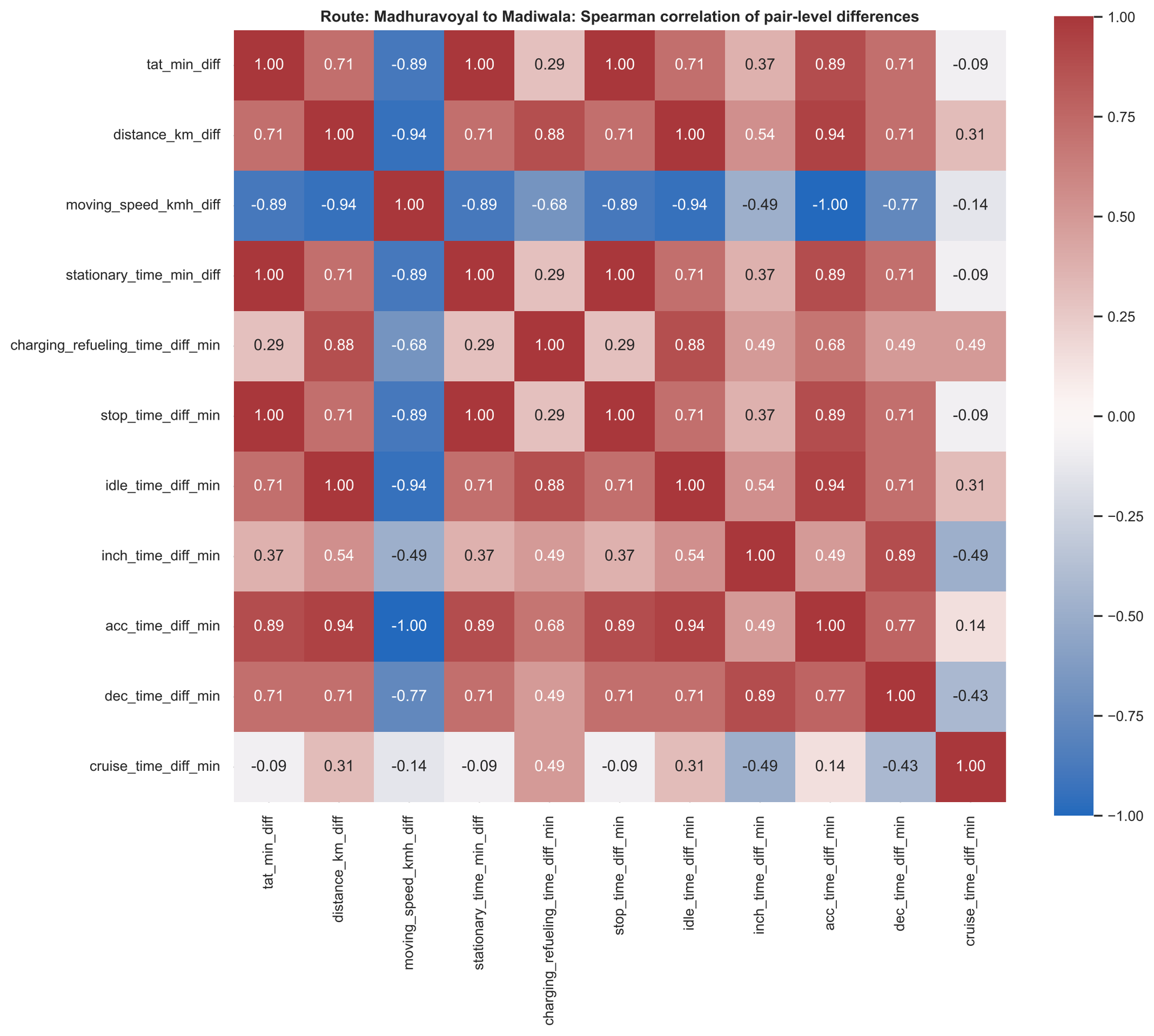

Route: Madhuravoyal to Madiwala: Distribution of TAT differences
Count
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−60
−40
−20
0
20
40
60
80
EV TAT − ICE TAT (min)

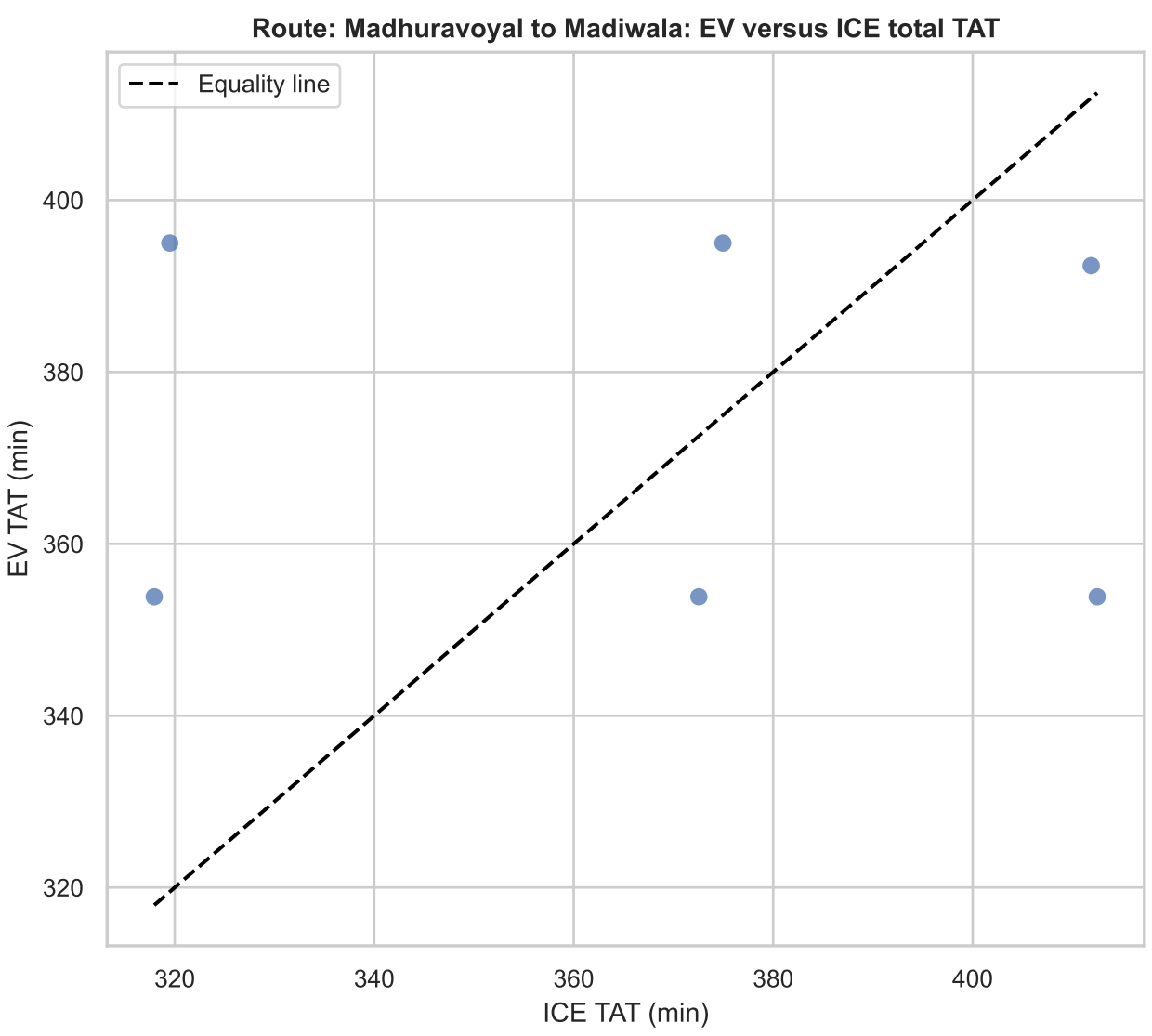

Route: Madhuravoyal to Madiwala: EV and ICE TAT distribution
Total TAT (min)
400
380
360
340
320
EV
ICE

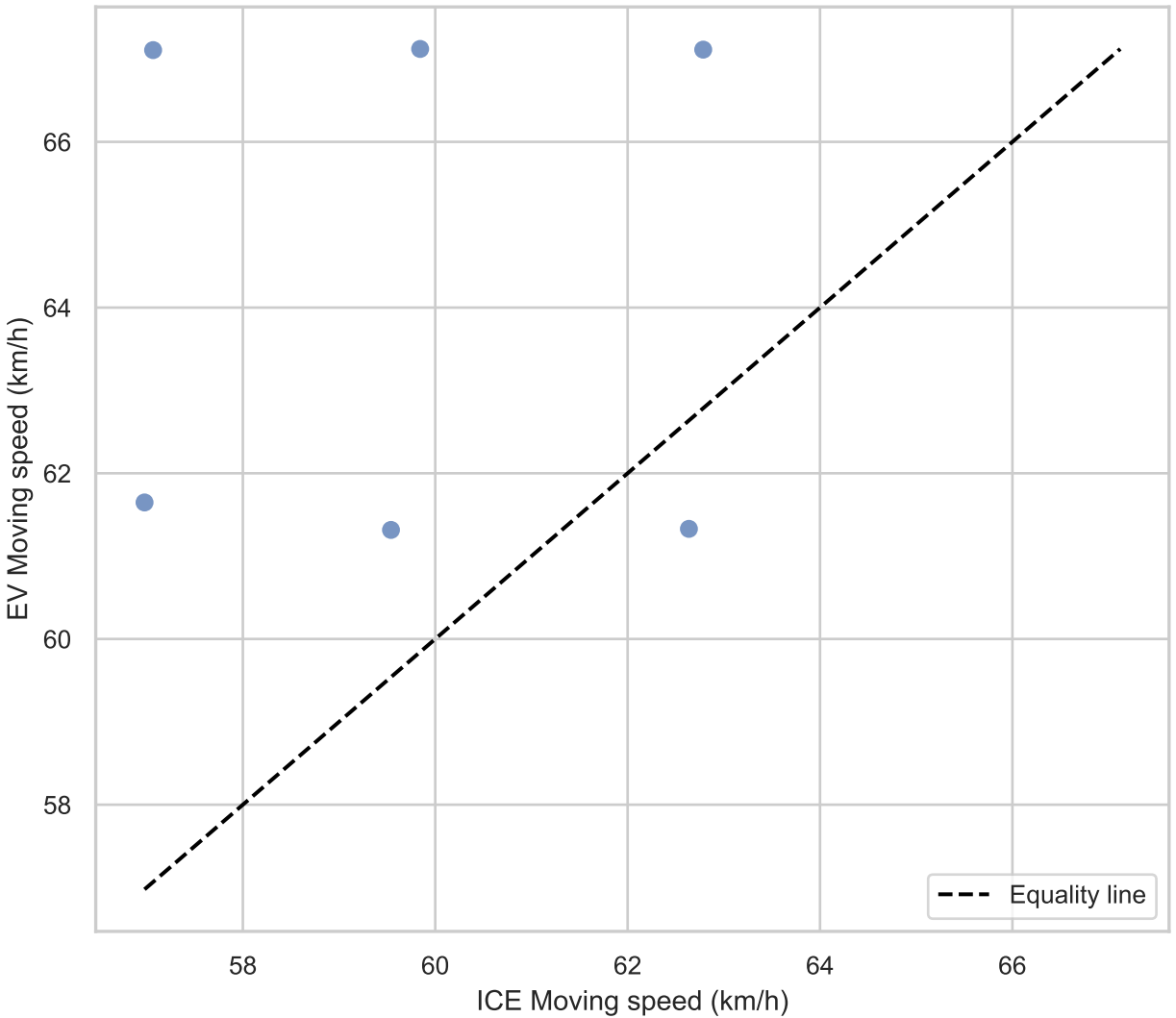

Route: Madhuravoyal to Madiwala: Pair-by-phase time difference
pair_id
PAIR_047
PAIR_039
PAIR_041
PAIR_038
PAIR_053
PAIR_040
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
40
20
0
−20
−40

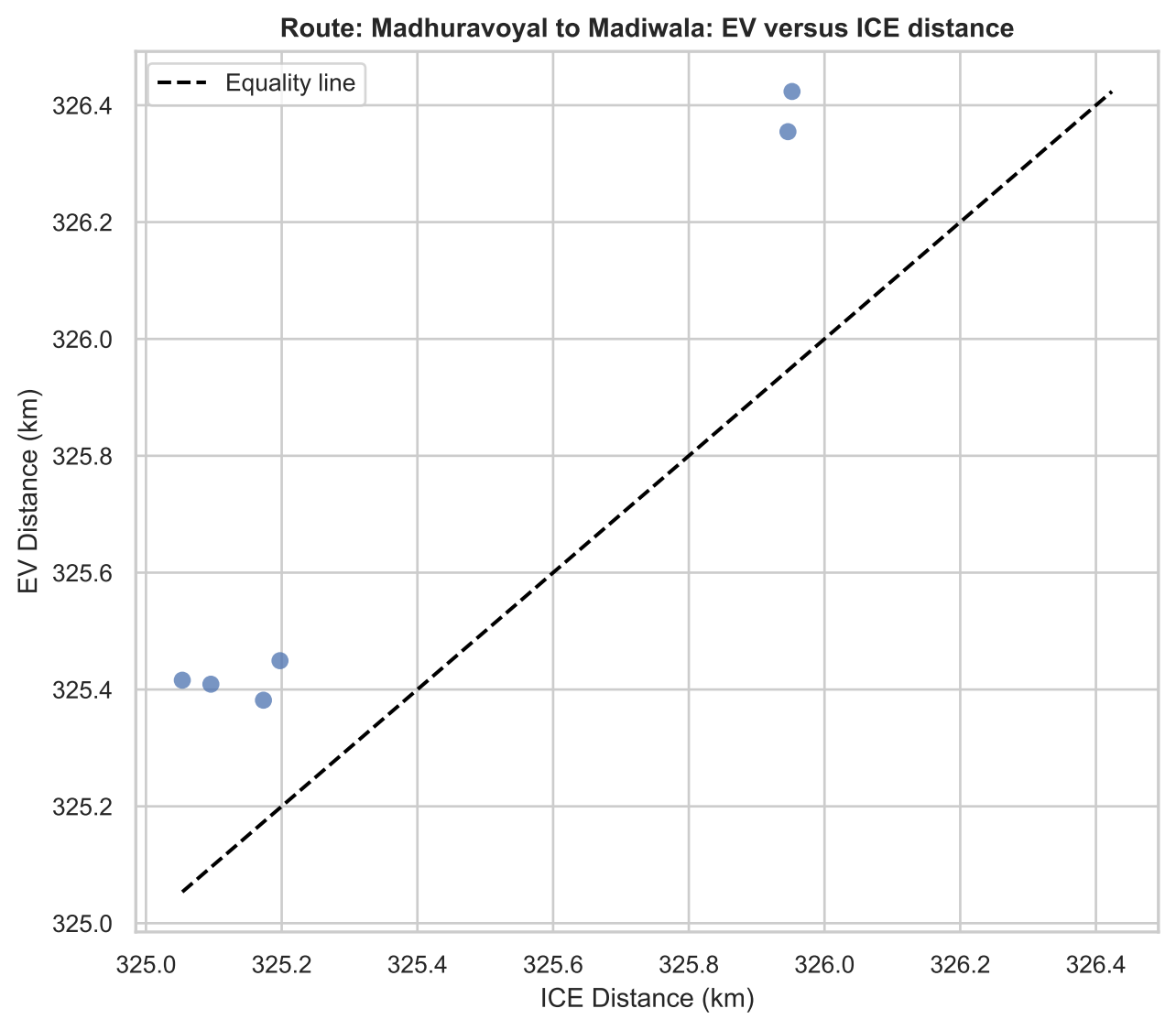

Route: Madhuravoyal to Madiwala: Mean phase contribution to TAT difference
Mean EV – ICE phase time (min)
40
20
0
−20
−40
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

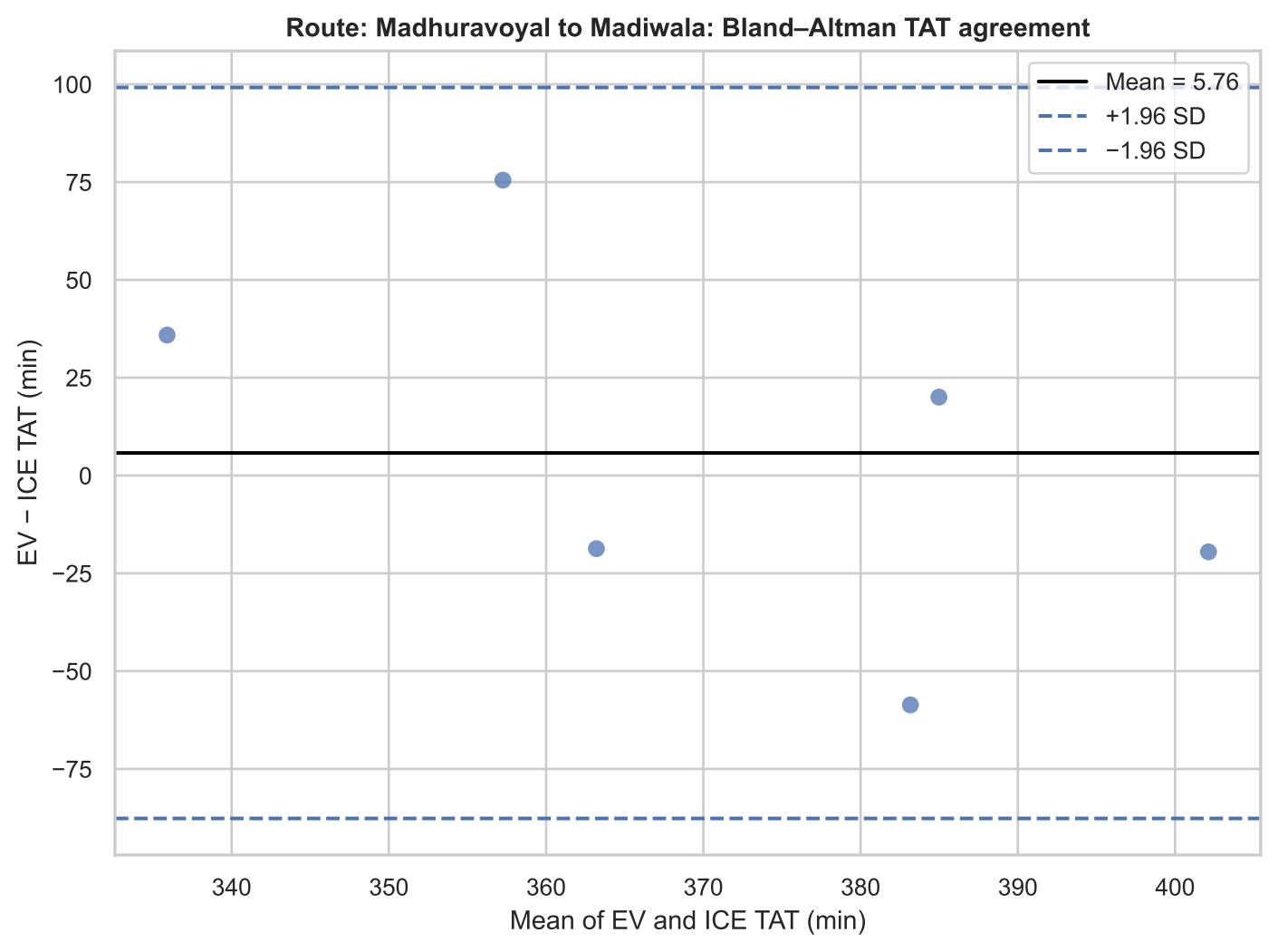

Route: Madhuravoyal to Madiwala: Pearson correlation of pair-level differences
tat_min_diff 1.00 0.83 -0.92 0.95 0.45 0.90 0.78 0.32 0.98 0.90 0.02
distance_km_diff 0.83 1.00 -0.98 0.61 0.87 0.52 0.95 0.36 0.90 0.72 0.28
moving_speed_kmh_diff -0.92 -0.98 1.00 -0.75 -0.76 -0.67 -0.94 -0.31 -0.97 -0.79 -0.24
stationary_time_min_diff 0.95 0.61 -0.75 1.00 0.14 0.99 0.55 0.29 0.87 0.89 -0.17
charging_refueling_time_diff_min 0.45 0.87 -0.76 0.14 1.00 0.03 0.84 0.30 0.58 0.36 0.43
stop_time_diff_min 0.90 0.52 -0.67 0.99 0.03 1.00 0.45 0.30 0.81 0.87 -0.26
idle_time_diff_min 0.78 0.95 -0.94 0.55 0.84 0.45 1.00 0.06 0.89 0.54 0.54
inch_time_diff_min 0.32 0.36 -0.31 0.29 0.30 0.30 0.06 1.00 0.21 0.68 -0.73
acc_time_diff_min 0.98 0.90 -0.97 0.87 0.58 0.81 0.89 0.21 1.00 0.81 0.22
dec_time_diff_min 0.90 0.72 -0.79 0.89 0.36 0.87 0.54 0.68 0.81 1.00 -0.39
cruise_time_diff_min 0.02 0.28 -0.24 -0.17 0.43 -0.26 0.54 -0.73 0.22 -0.39 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

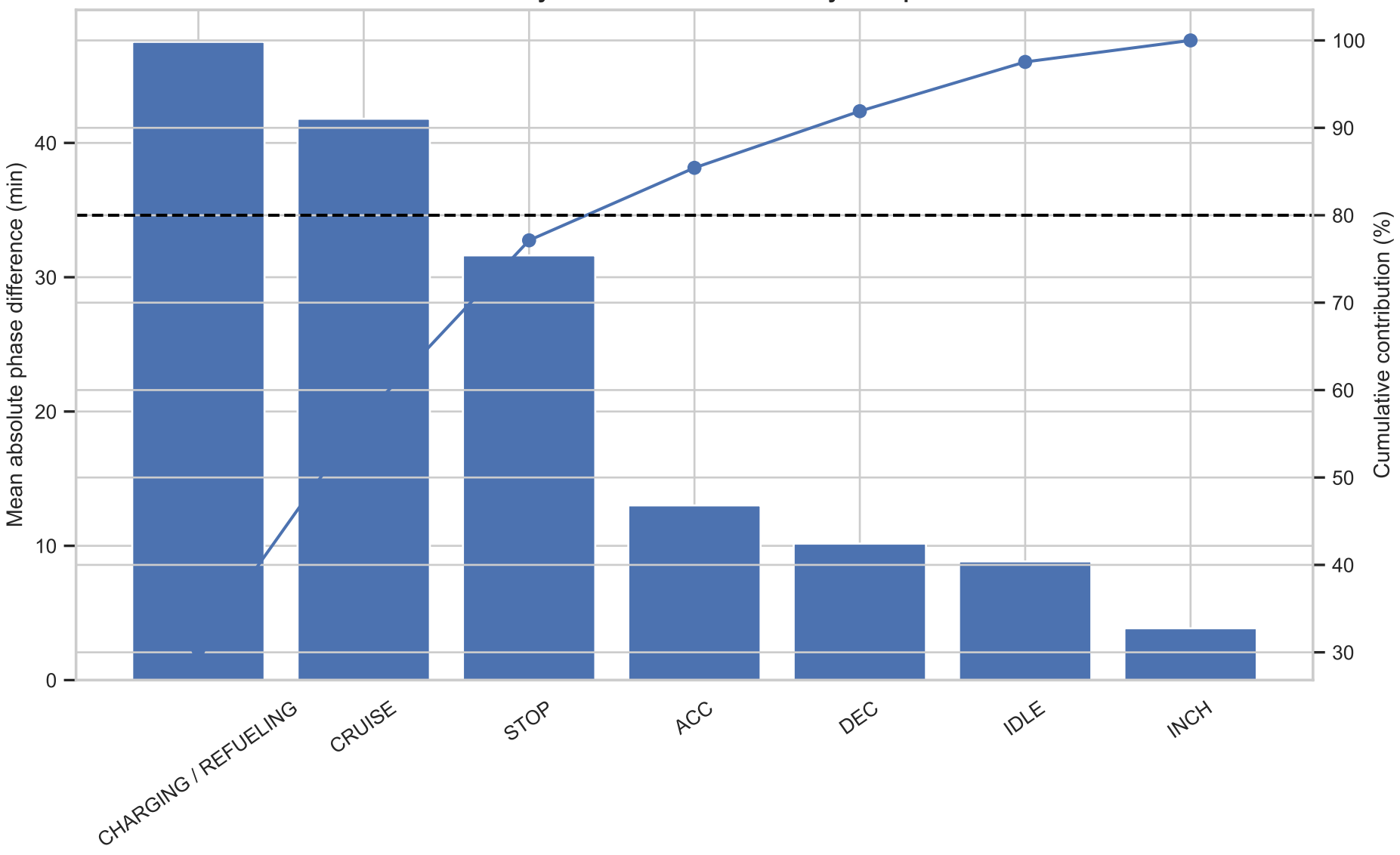

Route: Madhuravoyal to Madiwala: Spearman correlation of pair-level differences
tat_min_diff 1.00 0.71 -0.89 1.00 0.29 1.00 0.71 0.37 0.89 0.71 -0.09
distance_km_diff 0.71 1.00 -0.94 0.71 0.88 0.71 1.00 0.54 0.94 0.71 0.31
moving_speed_kmh_diff -0.89 -0.94 1.00 -0.89 -0.68 -0.89 -0.94 -0.49 -1.00 -0.77 -0.14
stationary_time_min_diff 1.00 0.71 -0.89 1.00 0.29 1.00 0.71 0.37 0.89 0.71 -0.09
charging_refueling_time_diff_min 0.29 0.88 -0.68 0.29 1.00 0.29 0.88 0.49 0.68 0.49 0.49
stop_time_diff_min 1.00 0.71 -0.89 1.00 0.29 1.00 0.71 0.37 0.89 0.71 -0.09
idle_time_diff_min 0.71 1.00 -0.94 0.71 0.88 0.71 1.00 0.54 0.94 0.71 0.31
inch_time_diff_min 0.37 0.54 -0.49 0.37 0.49 0.37 0.54 1.00 0.49 0.89 -0.49
acc_time_diff_min 0.89 0.94 -1.00 0.89 0.68 0.89 0.94 0.49 1.00 0.77 0.14
dec_time_diff_min 0.71 0.71 -0.77 0.71 0.49 0.71 0.71 0.89 0.77 1.00 -0.43
cruise_time_diff_min -0.09 0.31 -0.14 -0.09 0.49 -0.09 0.31 -0.49 0.14 -0.43 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

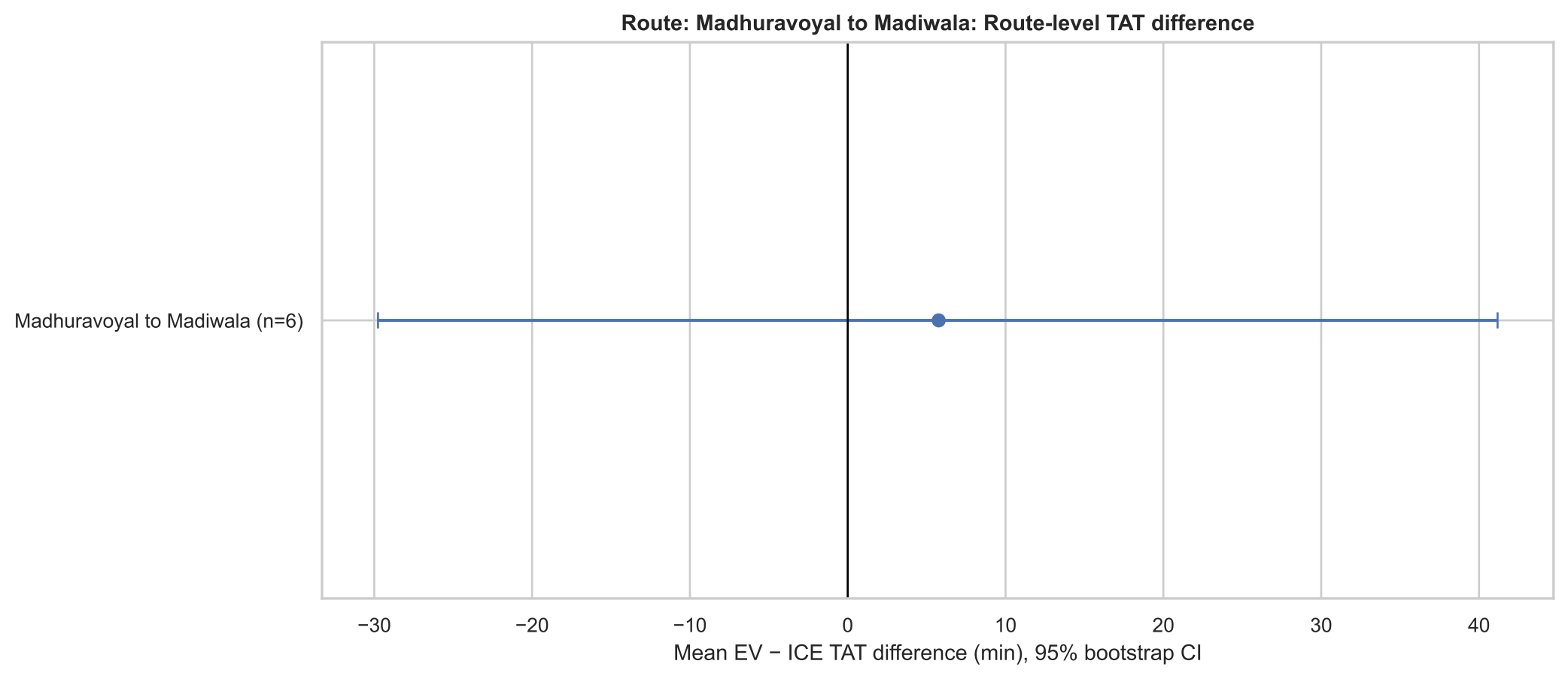

Route: Madhuravoyal to Madiwala: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
320
340
360
380
400

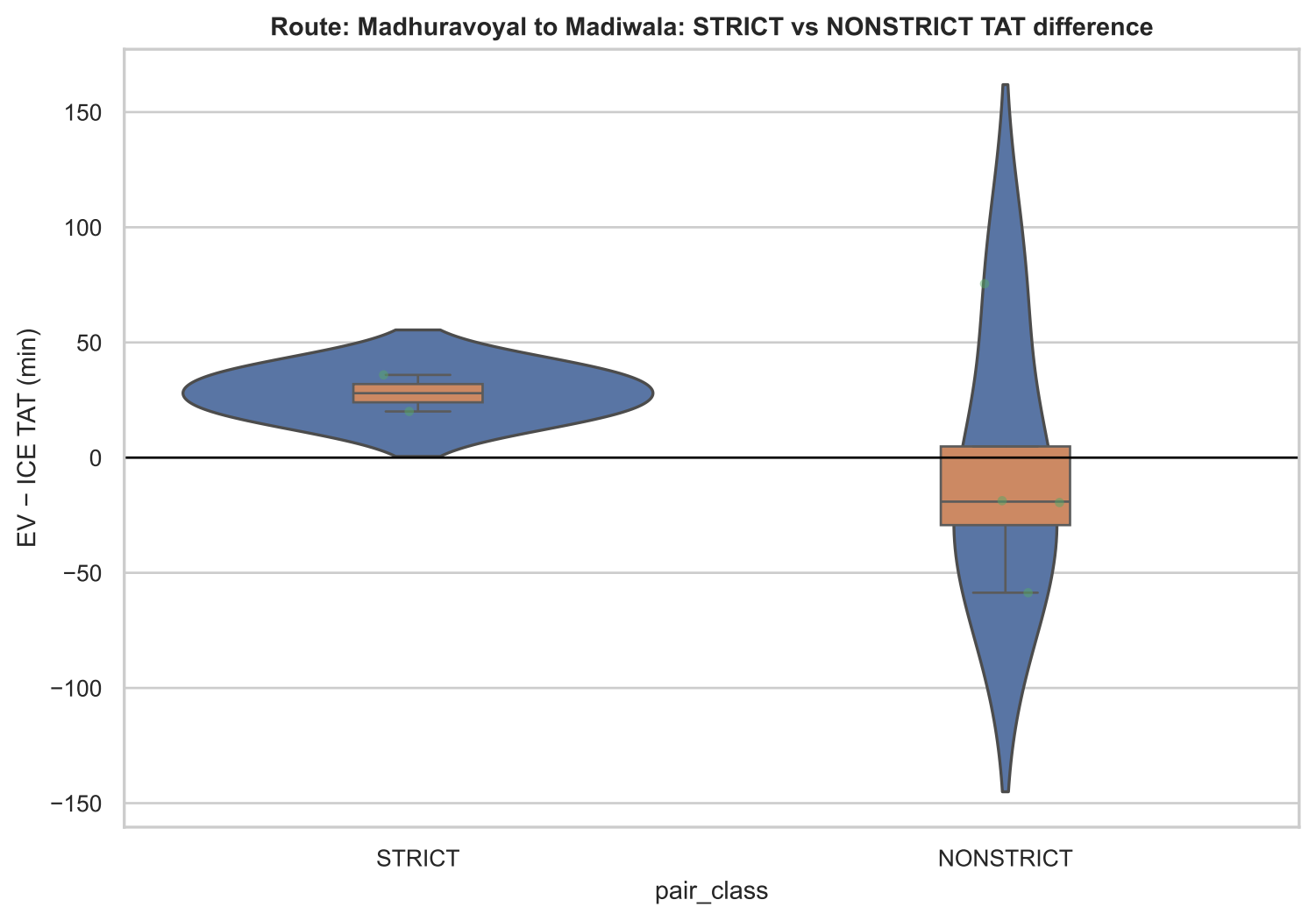

Route: Madhuravoyal to Madiwala: EV versus ICE moving speed
EV Moving speed (km/h)
ICE Moving speed (km/h)
66
64
62
60
58
58
60
62
64
66
Equality line

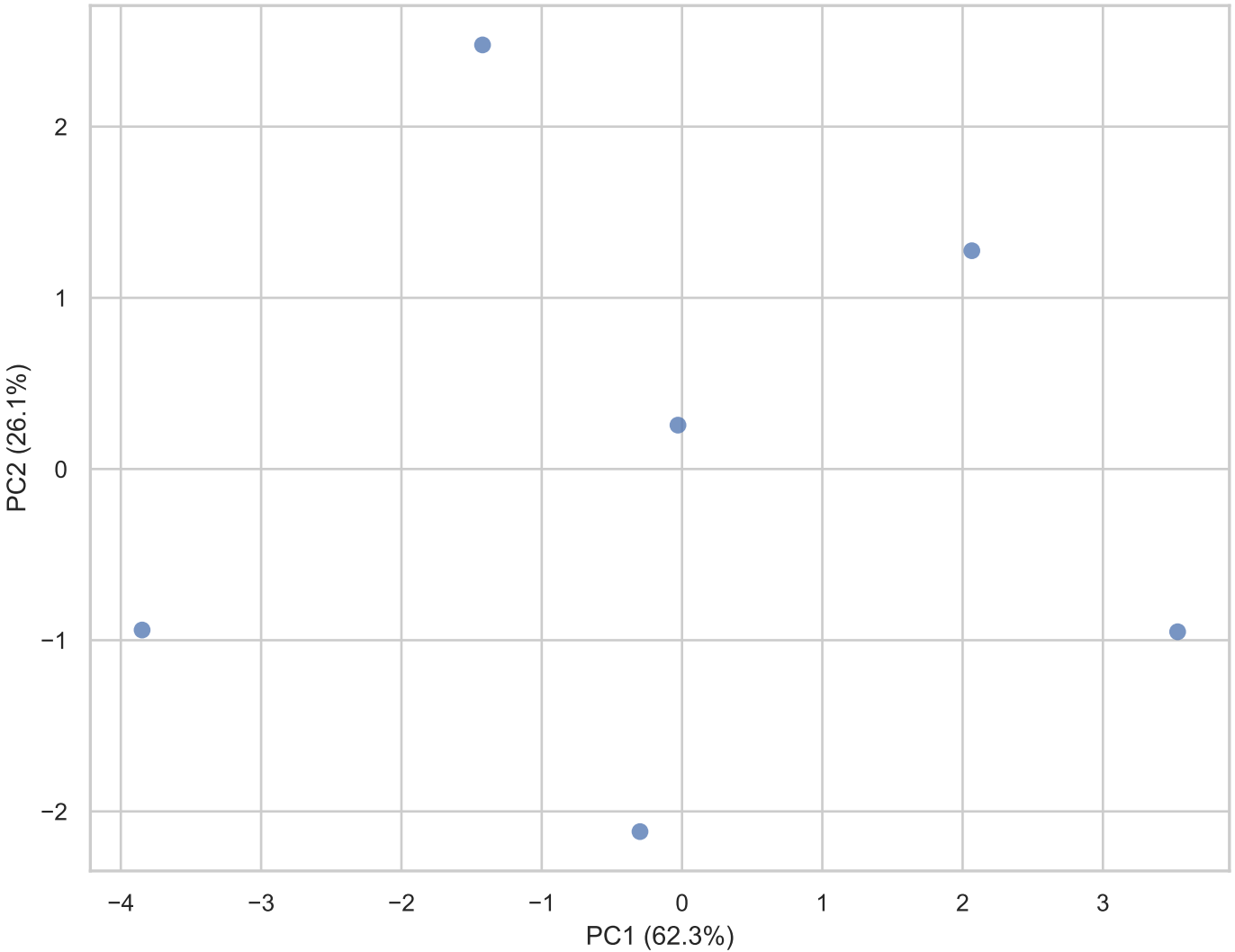

Route: Madhuravoyal to Madiwala: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
326.4
326.2
326.0
325.8
325.6
325.4
325.2
325.0
325.0
325.2
325.4
325.6
325.8
326.0
326.2
326.4

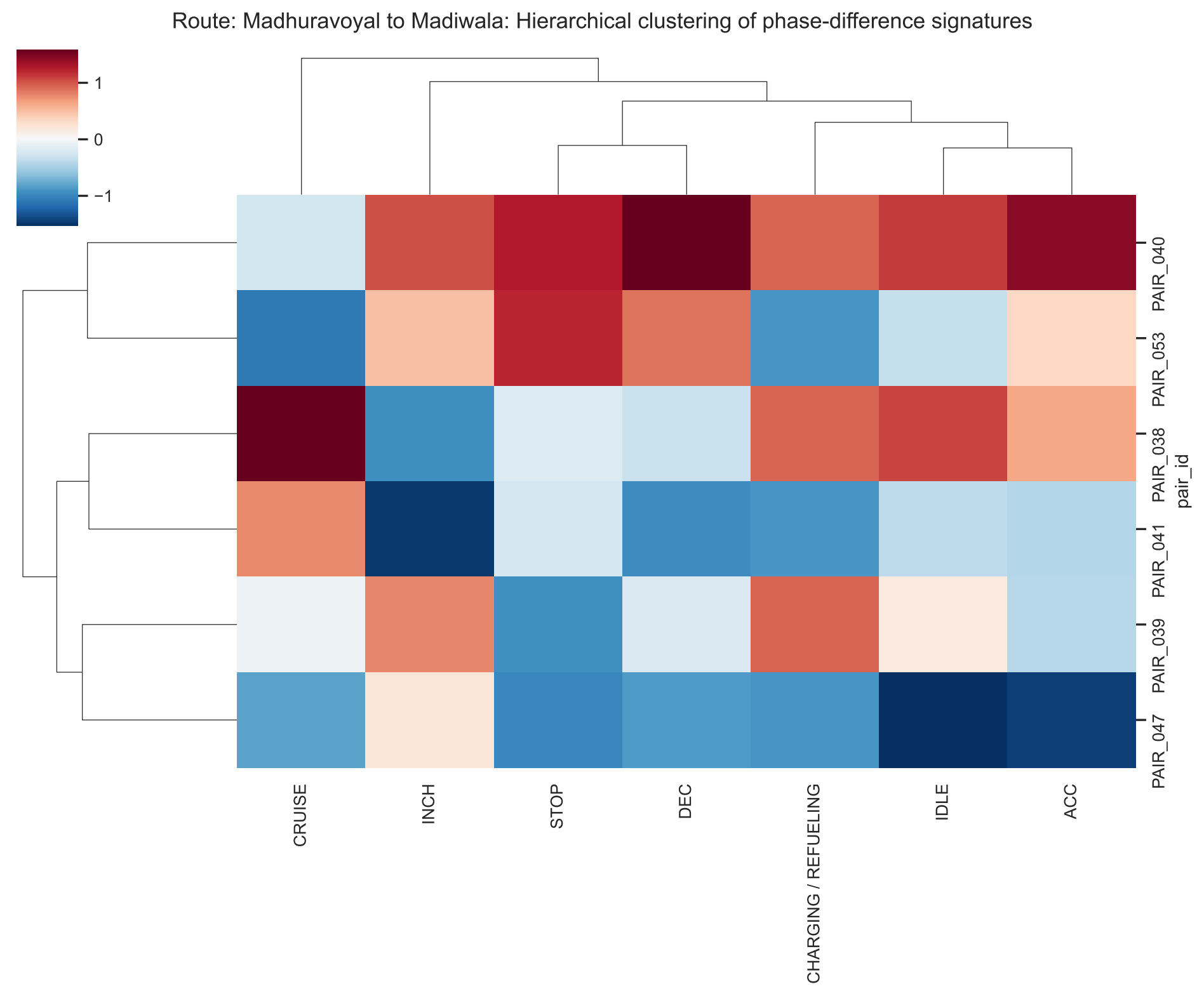

Route: Madhuravoyal to Madiwala: Bland–Altman TAT agreement
Mean = 5.76
+1.96 SD
−1.96 SD
100
75
50
25
0
−25
−50
−75
EV − ICE TAT (min)
340
350
360
370
380
390
400
Mean of EV and ICE TAT (min)


Route: Madhuravoyal to Madiwala: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
10
20
30
40
30
40
50
60
70
80
90
100
CHARGING / REFUELING
CRUISE
STOP
ACC
DEC
IDLE
INCH


Route: Madhuravoyal to Madiwala: Route-level TAT difference
Madhuravoyal to Madiwala (n=6)
−30
−20
−10
0
10
20
30
40
Mean EV − ICE TAT difference (min), 95% bootstrap CI


Route: Madhuravoyal to Madiwala: STRICT vs NONSTRICT TAT difference
150
100
50
0
−50
−100
−150
EV − ICE TAT (min)
STRICT
NONSTRICT
pair_class


Route: Madhuravoyal to Madiwala: PCA of EV−ICE operational differences
PC2 (26.1%)
PC1 (62.3%)
2
1
0
−1
−2
−4
−3
−2
−1
0
1
2
3


Route: Madhuravoyal to Madiwala: Hierarchical clustering of phase-difference signatures
1
0
−1
PAIR_040
PAIR_053
PAIR_038
PAIR_041
PAIR_039
PAIR_047
pair_id
CRUISE
INCH
STOP
DEC
CHARGING / REFUELING
IDLE
ACC